MISH 15
Lockout / Tagout (LOTO)

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Lockout / Tagout

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MH CONSTRUCTION INDUSTRIAL SAFETY AND HEALTH PROGRAM

Title

LOCK AND TAG

Number

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1.0 PURPOSE

This procedure implements OSHA/WISH standards minimum safety requirements for locking and tagging equipment and/or systems to protect personnel, property, and the environment from events that may be caused by the unexpected release of hazardous energy or materials.

Compliance with the Lock and Tag program is mandatory for all contractors and their lower tiers.

When working on the Hanford site this procedure does not apply.

MH Construction and subcontractors shall implement the DOE Lock and Tag program or DOE's Prime Contractors Lock and Tag Program while working on the DOE Sites.

2.0 RESPONSIBILITY

2.1 Affected Workers are responsible for compliance with this procedure.

2.2 Authorized Workers are responsible for:

 Compliance with this procedure

 Proper installation and removal of lockout/tagout devices in accordance with this procedure.

 Verify that hazardous energy and materials are adequately controlled.

2.3 Lock and Tag Administrators are responsible for:

 Ensuring that the technical review is performed to correctly describe the safe condition boundary.

 Evaluating the impact of the boundary with respect to personnel safety, facility operability, configuration control, public and environmental protection, and compliance with Technical Safety Requirements.

 Authorizing installation and removal of lockout/tagout devices.

 Authorizing the lifting or partial removal of lockout/tagouts.

NOTE: All or some of these responsibilities may be delegated (e.g., to an authorized worker designated by his or her management).

2.4 Project Management is responsible for:

 Ensuring affected workers, authorized workers, lock and tag administrators, and technical reviewers are trained on this procedure and the Hanford Site Lock and Tag Standard

 Ensuring appropriate technical review and pre-job planning is performed

 Ensuring proper documentation as required by this procedure and the Hanford Site Lock and Tag Standard

 Enforcing implementation of this procedure

 Ensuring required surveillances are performed

 Ensuring periodic inspection of the lock and tag program

3.0 DEFINITIONS

Affected Worker: A worker whose job requires him or her to be in an area where equipment or system(s) have been de-energized under the lock and tag procedure.

Authorized By: The section of lock and tag documentation that specifies the lock and tag administrator authorizing installation or removal of a tag.

Authorized Worker: A worker who has completed the required training and implements the lock and tag procedure by installing and removing lockout and tagout devices. An authorized worker may also be an “affected worker”. See also “principal authorized worker”.

Boundary: The limits of a given lockout/tagout as determined by those components that are configured to provide a safe condition where the work is to be performed.

Controlling Organization: The organization or individual responsible for the operation of the building, utility, facility, system, or equipment associated with the work to be performed.

Energy Source: Any source of hazardous energy or materials. Sources include, but are not limited to: electrical, mechanical, hydraulic, pneumatic, chemical, radiation, and thermal energies, as well as various forms of potential energy, such as that stored in springs, compressed gases, or in suspended objects (gravitational).

Installation: The process of using the lockout/tagout procedure to place a system or component in the proper configuration by applying lock(s) and tag(s) only, completing the required documentation, and notifying affected employees.

Isolating Device: A device that prevents the transmission or release of hazardous energy or hazardous materials. Examples include, but are not limited to: restraint blocks, electrical circuit breakers, disconnect switches, slide gates, slip blinds, or line valves. For lock and tag purposes, isolating devices that provide visible indication of the device's position are desirable.

Lock and Tag Administrator: The individual(s) responsible for implementation of the lock and tag program as designated by each contractor.

Locking Device (or Lock): A device that holds an isolating device in the required position for the protection of personnel and/or equipment.

Lockout: Installation of locking devices on the isolating devices to ensure that work can be performed safely. The locking devices ensure that the isolating devices (and the equipment or system(s) they isolate and/or control) cannot be operated until the locking device is removed.

Overlock/Overtag: Locks and tags that are installed by an authorized worker over a controlling organization's danger tag.

Personal Locking Device: A lockout device assigned to a specific worker and used in accordance with the requirements of 7.a.(10). A personal locking device may also be used as an overlock when used with an overtag.

Principal Authorized Worker: An authorized worker responsible for implementing the lock and tag procedure for a group of authorized workers.

Safe Condition: The condition established to permit work to be done within a lockout/tagout boundary, attained by isolating the work area from all sources of energy and/or hazardous materials outside the boundary and removing or securing all sources of potential energy inside the boundary.

Safe Condition Check: The inspection or test of a system or component performed to ensure that the energy or hazardous materials are adequately controlled to prevent injury or accident.

Tag: A caution or danger tag.

Caution Tag: A tagout device used for the protection of equipment or property, which are sometimes used to provide information or instruction.

Danger Tag: A tagout device used to establish safe work area boundaries and prevent the operation of any component or equipment when operation could cause personnel injury or death.

Tag Disposal: The process of destroying, cutting, or defacing tags to prevent reuse.

Tagout: The installation of a tagout device on an isolating device.

Tagout Device: A prominent warning device (such as a tag and means of attachment) that can be securely fastened to an isolating deice to indicate that the isolating device and equipment being controlled by it may not be operated (outside of the specification on the tag, for caution tags) until the tagout device is removed.

Technical Review: A review of the lockout/tag out boundary performed to ensure and adequate boundary is established. The review is performed using available information such as controlled drawings, documents, or field walk down.

Verification: Confirmation or checking of the lockout/tag out installation.

4.0 GENERAL REQUIREMENTS

4.1 GENERAL: This procedure serves to implement the requirements of the following standards:

 29 CRF 1910.147 (OSHA Lockout/Tagout Standards)

4.1.1 All employees of MH Construction and subcontractors under the control of MH Construction shall comply with this procedure. When working in another contractor's facilities, workers shall comply with facility specific work control requirements.

4.1.2 Only authorized workers or principal authorized workers (PAW) shall install or verify locks and tags under this procedure. (See definitions.)

4.1.3 When major replacement, repair, renovation, or modification of machines, systems, or equipment is performed, and whenever new machines or equipment are installed, isolating devices for such machines or equipment shall be designed to accept a lockout device.

4.1.4 If an isolating device is incapable of being locked, a tagout device shall be used. The level of protection shall be equal to that of a lockout, and the tagout device shall be installed to clearly indicate that the operation or movement of the isolating device from the "safe" or "off" position is prohibited. Examples of additional safety measures that may be taken to provide a level of protection equivalent to that of a lock are as follows:

 Remove an isolating circuit element.

 Block control of switches.

 Open extra disconnects in a circuit.

 Remove fuses.

 Remove a valve handle to reduce the likelihood that a system may become inadvertently energized.

4.1.5 When a locking device cannot be applied along with a danger tag, the tag attachment device used shall be durable and provide service that is at least equivalent to a one piece, all-environment-tolerant cable tie with a minimum strength of not less than 50 pounds, i.e., a Thomas

& Betts #TY523MX or the equivalent. At no time should any metallic attachment device be used.

4.1.6 No one shall operate any device on which a danger tag is hung.

4.1.7 No one shall operate equipment on which a caution tag is installed beyond the limits specified on the tag.

4.1.8 No one shall authorize another person to violate a lockout/tagout that is in place, except as specified in paragraphs 5.6.2 and 5.7.

4.1.9 Before an authorized or affected employee turns off a machine or equipment, the authorized employee shall have knowledge of the type & magnitude of the energy, the hazards of the energy to be controlled, & the methods or means to control the energy.

4.1.10 The machine or equipment shall be turned off or shutdown using the procedures established for the machine or equipment. An orderly shutdown must be utilized to avoid any additional or increased hazard(s) to employees as a result of the equipment stoppage.

4.1.11 When the following conditions are met, a personal locking device may be used by itself to establish a safe condition. In this case, the safe condition check shall be the only verification required. A lockout/tagout logbook entry shall not be required unless the lockout extends beyond a single shift and;

 The machine or equipment has no potential for stored or residual energy or re-accumulation of stored energy after shut down which could endanger employees

 The machine or equipment has a single energy source which can be readily identified and isolated

 The isolation and locking out of that energy source will completely de-energize and deactivate the machine or equipment

 The machine or equipment is isolated from that energy source and locked out during servicing or maintenance

 A single lockout device will achieve a locked out condition

 The lockout device is under the exclusive control of the authorized worker performing the servicing or maintenance

 The servicing or maintenance does not create hazards for other employees.

4.1.12 The controlling organization shall ensure procedures are in place to notify affected workers of the lockout/tagout.

4.2 INTERFACES: All contractor and subcontractor workers shall comply with this standard. When working in another contractor’s facilities, workers must comply with facility specific work control requirements.

4.2.1 Interfacing with Controlling Organization

4.2.1.1 The controlling organization establishes the boundaries of the lockout/tagout and the application of the Lock and Tag Standard.

4.2.1.2 Other organizations or individuals performing work on the equipment or system shall apply locks and/or tags in addition to those installed by the controlling organization. Locks and tags installed by the controlling organization will not be removed until any additional locks and/or tags have been removed by the organizations or individuals that installed the additional locks and/or tags.

4.2.1.3 When working in another contractor's facility, the authorized worker or the PAW coordinates the installation or removal of lockout/tagout devices with the Contractor's person in charge.

4.2.2 Interfacing with the Electrical Utilities Organization (EUO)

4.2.2.1 EUO may use their tags to overtag a danger tag

4.2.2.2 EUO provides tags and electrical clearances as a means of controlling boundary isolations

4.2.2.3 Workers may overtag an EUO tag or clearance with their own lock and/or tag. EUO is contacted to assist with this interface

4.2.3 Interfacing with Closed Fire Protection Valve Tags: A closed fire protection valve tag is used to notify management, the Fire Department, and the Contractor fire protection organization when any valve controlling the flow of water to a fire protection system is closed. A lockout/tagout shall be used in addition to a closed fire protection valve tag if required for personnel safety.

4.3 SAFE CONDITIONS AND LOCKOUT/TAGOUT BOUNDARIES: This section provides requirements which shall be met to ensure safe conditions are established when specifying lockout/tagout boundaries for the following hazard types:

4.3.1 Electrical Energy: This section applies to situations where the potential for personnel contact with electrical parts is the hazard. Requirements applicable to other hazards associated with electrically driven equipment (e.g. rotating or moving equipment) are provided in other sections.

This section does not apply to work on cord-and-plug-connected electrical equipment for which exposure to the hazards of unexpected energization or start-up of the equipment is controlled by the unplugging of the equipment from the energy source and by the plug being under the exclusive control of the employee performing the servicing or maintenance. The plug is under the exclusive control of the employee if it is physically in the possession of the employee, or in arms reach and inline of sight of the employee, or if the employee has affixed a lockout/tagout device on the plug.

4.3.1.1 During the lockout/tagout or prior to start of work, test equipment shall be used to test the circuit elements and electrical parts of equipment to which employees will be exposed and verify that the circuit elements and equipment parts are de-energized. I f the circuit to be tested is over 600 volts, nominal, the test equipment shall be checked for proper operation immediately before and after the test.

4.3.1.2 At least one of the following steps shall be performed to verify that equipment has been de-energized:

 Attempting to operate the equipment controls to demonstrate it cannot be started

 Checking the position indicators on electrical isolation devices

 Verifying that all disconnects are open.

CAUTION: Do not attempt to operate any controls tagged with a danger tag.

4.3.1.3 Work on energized parts shall be conducted only as permitted by MISH 29, subsections 5.2 and 5.3.

4.3.2 Rotating or Moving Equipment

CAUTION: Do not attempt to operate any controls tagged with a danger tag.

4.3.2.1 After the lockout/tagout is installed, attempt to operate the equipment using the operating controls, or otherwise verify that the equipment cannot be started.

4.3.2.2 Verify that the equipment is de-energized by checking the position indicators on electrical isolation devices, visually verifying that any disconnects are open, or performing a voltage check to ensure that there is no power available.

4.3.2.3 Where isolation from an energy source does not eliminate the potential for hazardous movement of equipment, the equipment shall be blocked or otherwise secured to prevent such movement.

4.3.3 Fluid Systems (Liquid or Gas)

4.3.3.1 Less than 200 F and 500 psig

 Systems and components that normally operate at temperatures and pressures above ambient shall be vented and, if possible, drained.

 Whenever possible, a danger tag shall be installed on an open atmospheric drain and/or vent between the equipment to de-pressurize the equipment and to accommodate thermal expansion or contraction.

 If a normal de-pressurization path cannot be provided within the work boundary, other methods shall be used to ensure that the system or component is adequately de-pressurized and drained such as loosening the fasteners on flanged connections or valve bonnets, removing instrument tubing, etc.

4.3.3.2 Greater than 200F or 500 psig: In addition to the requirements of subparagraph 4.3.3.1, the following shall apply:

4.3.3.2.1 At least two shutoff valves, in series, shall be used to provide isolation from fluid systems at greater than 200F or 500 psig (“two-valve protection”). The requirements for two-valve protection shall apply to all paths from which the fluid could cross the work boundary.

4.3.3.2.2 If two-valve protection is required but cannot be obtained, management may authorize the use of single-valve isolation provided:

 Alternate isolation devices (i.e., blank flanges, blocks, or freeze seals) have been considered and determined to be unfeasible or impracticable.

 Management approval of the single-valve isolation is documented; and

 The integrity of the single isolation valve is verified by venting or draining the portion of the system to be worked on and observing for leakage for at least 15 minutes to verify positive valve closure and leak tightness prior to starting work.

4.3.4 Engine-Driven Equipment (except motor vehicles)

4.3.4.1 The battery or other source of power shall be disconnected.

4.3.4.2 One or more essential operating parts, (e.g., coil wire, rotor, etc.) shall be removed or disconnected.

4.3.4.3 In the case of well drilling rigs equipped with ignition switches, subparagraphs 4.3.4.1 and 4.3.4.2 are not required. Rather, the individual performing the work shall remove the ignition key from the switch, place a completed “DANGER” tag at the ignition switch, and place the key in his/her pocket so that it will not be accessible to any other person(s).

4.3.4.4 It is recognized that well drilling rigs present unique and complex lock and tag requirements. Therefore, additional lock and tag requirements relative to well drilling rigs shall be identified and managed in accordance with a Job Safety Analysis (JSA) developed specifically for drilling rigs and meeting the intent of Hanford Site Lock and Tag Standard.

4.3.5 Valve Practices

4.3.5.1 Pneumatically or electrically operated valves that fail open shall not e considered shut for isolation purposes unless the valves’ supplies are isolated and a jacking device or gag is installed to shut or keep the valve shut.

4.3.5.2 Pneumatically or electrically operated valves that fail shut shall not be considered shut for isolation purposes unless the valves’ supplies are isolated and tagged out and the valves are verified as being shut.

4.3.5.3 The energy source to pneumatically or electrically operated valves used, as isolation devices shall be isolated and locked and/or tagged out once the valve is in the position required by the lockout/tagout documentation (drawings, plans, PCIs, etc.).

4.3.5.4 All local and remote pneumatic and electric valve operators shall be tagged when the valve is used as a system isolation boundary point.

4.3.5.5 Regulators and check valves shall not be used as isolation boundary valves unless the valve is mechanically restrained in the required position with a gagging device designed for that application.

4.3.5.6 Additional isolations shall be considered and specified as necessary to ensure protection when working on valve motor actuators with manual overrides, springs, or other operating mechanisms.

4.3.6 Stored or Residual Energy: Following the application of lockout/tagout devices to energy isolating devices, all potentially hazardous stored or residual energy shall be relieved, disconnected, restrained, and otherwise rendered safe.

4.4 TAGS AND LOCKS

4.4.1 Only danger and caution tags shall be used to indicate isolated systems or components, identify hazardous conditions or equipment associated with the unexpected release of hazardous energy or materials, or provide instructions to prevent injury to personnel or protect facility and equipment

4.4.2 Danger tags shall be used to establish safe work area boundaries and prevent the operation of any component or equipment when operation could cause personnel injury or death.

4.4.3 Caution tags may be used to identify limiting conditions for equipment operation, provide information for locked out equipment, protect property or the environment, or serve as an aid in procedural compliance. A caution tag shall not be used for personnel protection.

4.4.4 Designated lockout devices shall only be used for the control of hazardous energy sources or hazardous materials. Lockout devices shall be identified (by a color, label, tag, or other marking) and keyed to a specific individual or group (e.g., millwrights, power operators, etc.).

4.4.5 Tags and their means of attachment shall be capable of withstanding the environment to which they are exposed for the maximum period of time that exposure is expected.

4.4.6 Tags shall be legible and understandable with all spaces properly filled out.

4.4.7 Lockout devices shall be substantial enough to prevent removal without the use of excessive force or unusual techniques such as the use of bolt cutters or other metal cutting tools.

4.5 LOCKOUT/TAGOUT LOGBOOK: Each Contractor shall establish a Lockout/Tagout Logbook to provide a summary of information about lockouts and tagouts. Due to the diversity of the activities and the strong tie of the lock and tag program to the work control process, it is not practical to specify the format of the logbooks. However, the logbooks shall contain the following information, as a minimum (see attachment 15-A):

 Unique sequential number Type of tag Equipment or component affected Date installed Reason for installation Person authorizing installation Date removed Person authorizing removal Surveillances (may be a separate record).

4.6 LOCKOUT/TAGOUT SURVEILLANCE

4.6.1 Installed lockouts/tagouts shall be surveyed at least quarterly. The above frequency shall not be required for locks and tags located in radiological control zones or hazardous chemical control zones that pose unusual safety risks; however, if the hazard is reduced to an acceptable level, surveillance shall be performed.

4.6.2 An alternative surveillance schedule may be adopted for locks and tags that are projected to be active for one year or longer. These may be surveyed less frequently that once quarterly, but shall be surveyed at least once every six months.

4.6.3 The surveillance shall include proper placement of locks and tags, evidence of correct implementation of this standard, and an evaluation of the continuing need for the lockout/tagout. The surveillance, including the deficiencies found and corrective actions(s) taken, shall be documented and the documentation shall be available for review.

4.7 PERIODIC INSPECTION: A periodic inspection of the lockout/tagout procedure shall be conducted at least annually to ensure that the requirements of the procedure are being followed.

4.7.1 The periodic inspection shall be performed by an authorized worker who is not assigned responsibilities associated with the activities being inspected.

4.7.2 The periodic inspection shall be designed to correct any deviations or inadequacies observed.

4.7.3 The key elements of the periodic inspection shall include:

• Isolation

• Equipment shutdown

• Lock and tag placement

• Verifying equipment deactivation

• Equipment startup

• Documentation (tags, logbooks, etc.)

4.7.4 The periodic inspection, including the deficiencies found and corrective actions(s) taken, shall be documented and the documentation shall be available for review.

4.8 TRAINING: Lockout/Tagout training shall be provided as follows:

4.8.1 All employees shall receive lockout/tagout orientation, which shall consist of:

4.8.1.1 Instruction that the installation and removal of a lock and/or tag shall be conducted only with proper authorization as required by this procedure.

4.8.1.2 Instruction on limitations of tagout devices.

4.8.1.3 Emphasis on the disciplinary actions that may be taken for any employee who violates the lockout/tagout procedures included in this standard.

4.8.1.4 Any facility or site-specific procedures for lock and tag applicable to their work location.

4.8.2 Affected Workers shall be trained to the same criteria as all employees and shall also receive training consisting of:

4.8.2.1 The purpose and use of the Lockout/Tagout Procedure.

4.8.2.2 The applicable contractor or subcontractor implementation procedure.

4.8.3 Authorized Workers shall be trained to the same criteria as an affected worker and shall also receive training consisting of:

4.8.3.1 Methods to recognize the types and magnitude of hazards existing in their work area.

4.8.3.2 Means to control and isolate these hazards.

4.8.3.3 Practical hands-on training in lock and tag application and verification.

4.8.4 Technical Reviewers shall be trained to the same criteria as an authorized worker and possess, by virtue of other training and/or experience, additional facility specific knowledge applicable to the type of lock and tag issue to be addressed.

4.8.5 Lock and Tag Administrators shall be trained to the same criteria as an Authorized Worker and shall receive training in the administrative implementation of the lockout/tagout procedure.

4.8.6 Retraining Requirements

4.8.6.1 Appropriate retraining shall be provided to employees whenever there is a significant change in work assignment, a new hazard is identified, or there is a significant change in the Hanford Site Lock and Tag Standard or the applicable implementation procedure.

4.8.6.2 Authorized Workers shall be retrained on the Hanford Site Lock and Tag Standard or the applicable implementation procedure annually.

4.8.7 All training and/or retraining must be documented, signed & certified.

5.0 PROCEDURE

5.1 PREPARATION AND REVIEW OF LOCKOUTS/TAGOUTS

5.1.1 All isolating devices for affected equipment and systems are located and identified.

5.1.1.1 Available information (such as controlled drawings, documents, and/or a field walk down) is used to verify that energy sources or hazardous materials requiring isolation are adequately identified. Several sources may be involved with the isolation of the equipment or system

5.1.1.2 Electrical control circuits do not provide adequate protection to interrupt main power and are not used as an isolation point for establishing a lockout/tagout boundary (Although they may be tagged to protect the equipment they are insufficient to provide a safe condition and isolation boundary for personnel protection).

5.1.2 The lock and tag administrator:

• Ensures that the technical review is performed to correctly describe the safe condition boundary.

• Evaluates the impact of the boundary with respect to personnel safety, facility operability, configuration control, public and environmental protection, and compliance with technical safety requirements.

• Authorizes installation of lockout/tagout devices.

NOTE: All or some of these responsibilities may be delegated (e.g. to an authorized worker designated by his or her management).

5.1.3 The tag is obtained and completed by the authorized worker or PAW.

5.1.4 When the conditions for placement of the tag are met, the “Authorized By” block is completed. The “Authorized By” block may be completed by the lock and tag administrator, authorized worker, or PAW based on verbal direction of the authorizing person.

NOTE: The authorizing lock and tag administrator may be a Key Supervisor, Division Manager, or other responsible person who has completed lock and tag administrator training as defined in subsection 5.8.

5.1.5 The appropriate entries are made in the Lockout/Tagout Logbook, when applicable.

5.2 LOCKOUT/TAGOUT INSTALLATION

5.2.1 The Contractor provides lock and tag cabinets at locations in proximity to the lock and tag operations. The cabinets are controlled by the Lock and Tag Administrator. The Lock and Tag Administrator will usually be the supervisor but may be delegated (e.g. to an authorized worker or PAW designated by his or her management).

5.2.1.1 Each cabinet contains numbered locks and tags. Locks are individually keyed so that the key of one lock does not open any other lock.

5.2.1.2 The cabinets are locked. Cabinets are individually keyed so that the key of one cabinet does not open the lock of any other cabinet.

5.2.2 Lockout and tagout installation may be accomplished by individual authorized workers or as a group.

5.2.2.1 In group lock and tag situations, an individual is designated as the “Principal Authorized Worker” (PAW) for his/her group. Each group or craft designates its own PSW.

5.2.2.2 A PAW may be the manager, supervisor, foreman, or a member of the crew assigned to work on the system being isolated.

5.2.3 The component is placed in the position specified by the lockout/tagout. In some cases, the authorized worker or PAW may need to have this function performed by the controlling organization. In such cases, the authorized worker or PAW ensures that the controlling organization has identified and positioned the component properly.

5.2.4 The authorized worker or PAW:

• Obtains the lock and/or tag from the designated location.

• Completes the tag (and the logbook wherever applicable).

• Installs the lock and/or tag on the equipment or system identified.

In operations where more than one group is working on the equipment or system, each PAW installs a lock and tag.

5.2.4.1 If applicable, the controlling organization installs locks and/or tags first. The authorized worker or PAW installs his/her lock and/or tag after the controlling organization’s lock and/or tag.

5.2.4.2 Where more than one individual or group is working on equipment, components, or systems, each subsequent authorized worker has the option of installing his/her own lock and/or tag.

5.2.5 Whenever an isolating device is not designed to accept multiple locks, the first worker to place a lock on the isolating device installs a multi-lock device to ensure that the isolating device can accommodate additional locks.

5.2.6 If an isolating device is not designed to accept multiple locks and the procedure in paragraph 5.2.5 is not feasible, a principal lock is affixed to the isolating device and the key is placed in a lock box provided specifically for that purpose. Locks belonging to any additional groups/crafts are affixed to the lock box so that the key for the principal lock cannot be accessed until all locks have been cleared from the lock box.

5.2.7 Danger tags are not used without locks where an isolating device is capable of being locked. Workers do not apply multiple danger tags to a single lock.

5.2.8 The “installed by” block on the tag is completed.

5.3 SAFE CONDITION CHECK

5.3.1 A safe condition check is performed as specified in subsection 4.3

5.3.2 When application of lockout/tagout precludes performance of the safe condition check, this check may be performed before installing a tagout device provided there is positive means of ensuring that the safe condition will be maintained between the time that the check is completed and the lock and/or tag is installed.

5.4 VERIFICATION

5.4.1 The position and/or condition of the equipment or components covered by the lockout/tagout is verified.

5.4.2 The “Verified By” block on the tag is completed. At this point, no spaces on the tag should be left blank. Not applicable (NA) is entered in blocks that do not apply.

5.5 NOTIFICATION: The controlling organization is notified after the tag is placed.

5.6 LOCKOUT/TAGOUT REMOVAL

5.6.1 Overlock and/or Overtag Removal

5.6.1.1 If an overlock or overtag has been installed, the authorized worker removes his or her lock and/or tag once the work has been completed.

5.6.1.2 The controlling organization is notified that the overlock or overtag has been removed.

5.6.2 Lockout/Tagout Removal

5.6.2.1 Authorization to remove the final lock(s) and/or tag(s) is obtained prior to removal.

5.6.2.2 Prior to removing the final lock and/or tag, the authorized worker or PAW conducts a physical check of the area to ensure that all personnel working under the protection of the lock and/or tag are accounted for. This check ensures the following:

• All safeguards, shields, tools and equipment, etc. are removed from the area.

• All affected workers are removed to a safe location.

5.6.2.3 The lock and/or tag is then removed as follows:

• Tags and locks are removed by the worker(s) or group(s) originally installing them. When the appropriate worker or group is not available, the device(s) may be removed under the direction of the authorized worker’s or group’s management after all reasonable efforts to reach the worker or group have been taken. The original worker or group must be informed of the lock and tag removal upon returning to work.

• The controlling organization’s lock(s) and/or tags are the last to be removed.

NOTE: The following steps described in subparagraphs 5.6.2.4 through 5.6.2.6 may be completed out of sequence.

5.6.2.4 Lockout/Tagout Logbook entries are completed as required.

5.6.2.5 All used tags are destroyed. If tags were removed in a radiological control zone or hazardous chemical control zone, the authorized worker(s) or PAWs removing the tags ensure that the tags cannot be reused by cutting or defacing the tags and discarding them in accordance with the appropriate disposal procedures applicable to the area.

5.6.2.6 The system is restored in accordance with appropriate work control procedures.

5.6.2.7 When work is to be suspended, the overlock(s)/overtag(s) may be removed if appropriate. Removal of tags will be conducted in accordance with this subsection and as follows:

 The controlling organization's lock/tag is left in place if required for safety reasons during the work suspension.

 If the controlling organization's lock/tag will be removed, the overlock(s)/overtag(s) are removed first.

5.7 TRANSFERRING LOCKOUT/TAGOUT DEVICES: When lockout/tagout devices are transferred during shift or personnel changes, the transfer of activities are documented in accordance with specific procedures within an organization. Documentation includes:

• The names of the off-going and oncoming personnel.

• Date and time of the transfer.

• Special instructions (if applicable).

• Identification of locks/tags to be transferred.

• Name of the lock and tag administrator who authorized the transfer.

• Reason for the transfer (vacation, leave of absence, transfer, termination, etc.).

5.8 SPECIAL CASES

5.8.1 Testing or Troubleshooting

5.8.1.1 Facility specific procedures may prohibit lifting of tags for any reason. Those specific procedures are followed.

5.8.1.2 In other cases, temporary removal of lockout/tagout devices is discouraged. However, for situations in which lockout/tagout devices must be temporarily removed and the equipment energized, removal of lockout or tagout devices and the re-energization of the machine or equipment is allowed only during the limited time necessary for the testing or positioning of machines, equipment or components. The following sequence of operations is followed:

 The equipment is cleared of tools and materials All personnel are removed to a safe location. The lockout/tagout devices are removed and retained. After completion of the temporary re-energization, the authorized worker reinstalls the lockout/tagout.

NOTE:When the testing or troubleshooting results in a situation where the equipment is to be returned to service or is to be turned back to the controlling organization, de-energization and re-locking or re-tagging is not performed. Rather, final removal of the lock and tag is completed.

5.8.2 Missing, Mutilated, or Illegible Tags: If a tag is found to be mutilated or illegible or is determined to be missing, the following actions are taken:

 Verification that the lockout or tagout is still active and required.

 A replacement tag is obtained and completed.

• The next available sequential number in the lock and tag logbook is used for the replacement tag.

 The replacement of the tag and the number of the replacement tag is noted in the designated space of the Lockout/Tagout Logbook

 The tag is installed in accordance with sections 5.2 through 5.5 of this standard.

5.9 RECORD RETENTION: Logs, surveillance records, and periodic inspection records generated by this procedure shall be maintained for a minimum of six months.

6.0 RECORDS

Document Record Submittal Responsibility Record Retention Responsibility Logs, Surveillance Records, and Periodic Inspection Records NA Contractor

7.0 REFERENCES

29 CFR 1910.333 Safety Related Work Practices for Electrical Work

8.0 ATTACHMENTS

MISH 15-A Lockout/Tagout Sign-In/Sign-Out Log

MH CONSTRUCTION LOCK AND TAG LOG BOOK PAGE _____ OF _____

TAG # COMPONENT TAGGED LOCK REQUIRED? POSITION/CONDITION AUTHORIZED BY/DATE INSTALLED BY/DATE VERIFIED BY/DATE SAFE CONDITION CHECK BY/DATE REMOVAL APPROVAL BY/DATE REMOVAL BY/DATE

MISH 15 — Lockout / Tagout (LOTO)

Scan to view this section online, download the PDF, or access the full MH Construction Safety Manual.

https://www.mhc-gc.com/resources/safety-manual/section/lockout-tagout-loto